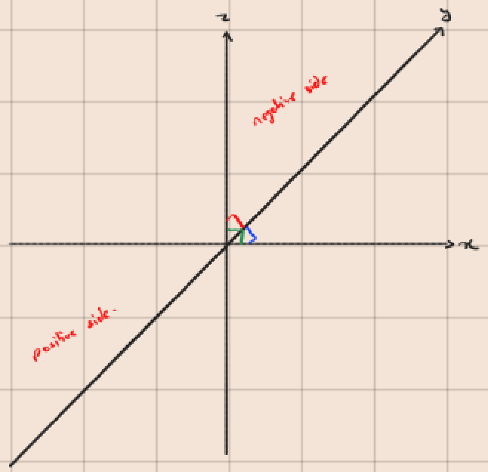
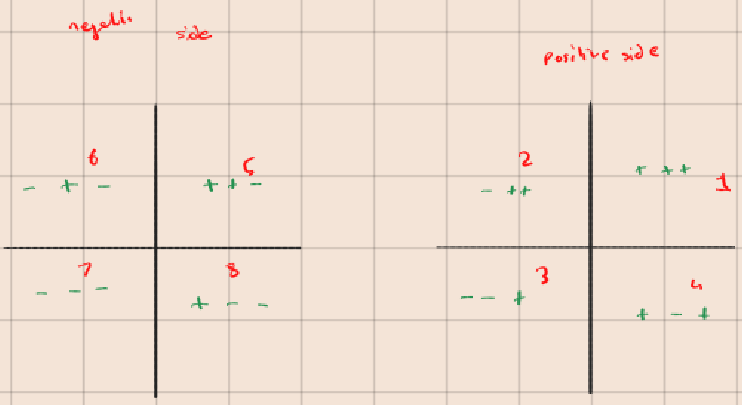
Coordinate system in 3D:

(x, y, z)



Octants

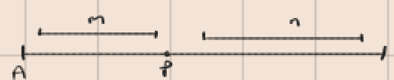
	x	y	z
1	+	+	+
2	−	+	+
3	−	−	+
4	+	−	+
5	+	+	−
6	−	+	−
7	−	−	−
8	+	−	−



Distance formula:

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 \; (z_2 - z_1)^2}$$

Section formula:



$$P = \left(\frac{m x_2 + n x_1}{m+n} , \frac{m y_2 + n y_1}{m+n} , \frac{m z_2 + n z_1}{m+n} \right)$$

Direction Cosine => DC (tells us the angles a lines make with x, y, z)

Direction Ratio => DR (tells us the direction of the line

DC is written as DC = $(\cos\alpha, \cos\beta, \cos\gamma)$ $\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$A(x_1, y_1, z_1)$ $B(x_2, y_2, z_2)$

DR = $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$

Relationship between DR and DC

If DR = (a, b, c)

then DC = $\left(\frac{a}{\sqrt{a^2+b^2+c^2}} + \frac{b}{\sqrt{a^2+b^2+c^2}} + \frac{c}{\sqrt{a^2+b^2+c^2}} \right)$

Q) $A(1,2,3)$ $B(4,6,9)$ find DR

Q) DR = $(2, -1, 2)$ find DC

Q) Verify that DC = $\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$ satisfy $l^2 + m^2 + n^2 = 1$

Vector equation of a straight line

1) Vector form

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

position vector

Point

Direction vector

2) Parametric form

point / $\vec{a}$ = (x_1, y_1, z_1)

direction vector / $\vec{b}$ = (a, b, c)

$$x = x_1 + \lambda a \quad , \quad y = y_2 + \lambda b \quad , \quad z = z_2 + \lambda c$$

3) Symmetric form

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Q) point $(2,-1,3)$ $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$

i) write the vector equation of a line Ans $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$

ii) Hence write it's parametric form Ans $x = 2 + \lambda$, $y = -1 - 2\lambda$, $z = 3 + 2\lambda$

Q) point $(1,2,3)$ $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$

i) write vector equation $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + \hat{j} - \hat{k})$

ii) Value of λ when student reaches the point $(5,4,1)$ $\lambda = 2$

iii) Is the point $(1,2,3)$ on the same path. yes $\lambda = 3$

Q) $A(1,2,3)$ $B(3,6,5)$

i) find equation

Q) $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{i} + 2\hat{j} - \hat{k})$

i) write equation in cartesian form

$x = 2 + \lambda 3$ $y = -1 + 2\lambda$ $z = 1 - 1\lambda$

ii) find (x, y, z) when $z = 0$ and point is on the line.